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Development and validation of the volatile correction model for PM_{10} – An empirical method for adjusting TEOM measurements for their loss of volatile particulate matter

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ABSTRACT

EU Directives stipulate that PM_{10} should be measured using the gravimetric reference method as laid out in EN12341 [CEN, 1998. Air Quality – Determination of the PM_{10} Fraction of Suspended Particulate Matter – Reference Method and Field Test Procedure to Demonstrate Reference Equivalence of Measurement Methods. European Committee for Standardisation], or an equivalent method as demonstrated using EC guidance [EC, 2005. Demonstration of Equivalence of Ambient Air Monitoring Methods. European Commission Working Group on Guidance for the Demonstration of Equivalence]. There is however a conflict between the requirement to measure PM_{10} using the gravimetric reference method and the need for rapid public reporting, and many member states, including the UK, rely on non-gravimetric techniques to measure PM_{10}. In the UK the majority of PM_{10} measurements are made using the Tapered Element Oscillating Microbalance (TEOM), which does not meet the equivalence criteria [Harrison, D., 2006. UK Equivalence Programme for Monitoring of Particulate Matter. Defra, London]. The implied need to upgrade or replace TEOMs with an equivalent automated measurement technique has significant cost implications. The model described in this paper was based on analysis of daily mean measurements of PM_{10} by the Filter Dynamics Measurement System (FDMS) and the TEOM at UK sites. It uses the FDMS measurement of the volatile component of PM_{10} (referred to here as FDMS purge) to correct for differences in the sensitivity to volatile PM_{10} between the TEOM and the EU gravimetric reference method. The model equation for the correction of TEOM PM_{10} measurements is: $TEOM_{VCM} = TEOM - 1.87$ FDMS purge due to the regional homogeneity of volatile PM, the FDMS purge concentration may be measured at a site distant to the TEOM, allowing the possibility of using a single FDMS instrument to correct PM_{10} measurements made by several TEOMs in a defined geographical area. The model was assessed against the criteria for the EC Working Group's Guidance for the Demonstration of Equivalence of Ambient Air Monitoring Methods [EC, 2005. Demonstration of Equivalence of Ambient Air Monitoring Methods. European Commission Working Group on Guidance for the Demonstration of Equivalence]. The model satisfies the equivalence criteria using remote FDMS purge measurements for distances up to 200 km (in 22 out of 23 data sets). These data provide strong evidence that the model is a viable tool for correcting measurements from TEOM instruments on the national and local government networks.

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1. Introduction

The UK's obligations regarding air quality are set out in EC Directive 2008/50/EC on ambient air quality and cleaner air for Europe. This directive sets Limit and Target Values for individual air pollutants along with data quality requirements with respect to uncertainty and data capture. It also stipulates that PM_{10} should be measured gravimetrically as laid out in EN12341 (CEN, 1998). The requirement to measure PM_{10} gravimetrically is at odds with the requirement for rapid public reporting due to the time between sampling, weighing and reporting the data, which can be up to 21–28 days after the sample was taken. There are also acknowledged positive and negative artefacts associated with the gravimetric method from the adsorption of organic gases onto the filter (Kirchstetter et al., 2001), the absorption and evaporation of semi-volatile organic and inorganic components (Chow et al., 2002) and incomplete removal of particle bound water (Price et al., 2003). For these reasons, many member states use automated PM_{10} measurement techniques.

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In the UK the majority of PM_{10} measurements are made using the TEOM, which has the widely acknowledged disadvantage of driving off semi-volatile material such as ammonium nitrate and organic aerosols (Ruppecht et al., 1992; Allen et al., 1997; Salter and Parsons, 1999; Soutar et al., 1999; Cyrys et al., 2001; Green et al., 2001; Charron et al., 2004). A 'correction' factor of 1.3 was therefore recommended in the UK for comparison of TEOM PM_{10} measurements with the EU Directive (DETR, 2000), however, this correction factor has its own limitations due to site and seasonal variability (Green et al., 2001).

Between 2004 and 2006, the UK Department for the Environment, Food and Rural Affairs (Defra) undertook a set of field trials to determine the equivalence of several automated and non-automated PM_{10} and $PM_{2.5}$ measurement techniques (Harrison, 2006) using the test methodology prescribed by the EC (EC, 2005). Harrison made some practical interpretation of the EC test methodology; these were felt to be necessary for the realistic application of the methodology. Several instruments, including the FDMS, proved equivalent to the European PM_{10} gravimetric reference method. Importantly, the TEOM did not meet the equivalence criteria (with the 1.3 or any other correction factor) and is therefore not suitable for reporting PM_{10} against the EU Limit Values. The implied need to upgrade or replace TEOMs with an equivalent automated measurement technique has significant cost implications for local and national governments.

Analysis of measurements of PM_{10} made using FDMS and TEOM measurements in London during 2004 and 2005 proposed a relationship which utilised FDMS purge measurements to maintain 'continuity' between measurements of PM_{10} made using the FDMS and those made using the TEOM (Green, 2004; Green and Fuller, 2006a,b). Given that a relationship could be derived to achieve continuity between TEOM and FDMS, and that the FDMS is equivalent to the EU gravimetric reference method, it should be possible to create a model to correct TEOM measurements so that they can meet the equivalence criteria.

2. Measurement methods

Measurements of PM_{10} were made at a range of locations using the different methodologies described below between 1st January 2004 and 19th January 2006; these sites are detailed in Table 1. These measurements formed part of the UK Equivalence Programme, the UK Automatic Urban and Rural Network (AURN) and the London Air Quality Network (LAQN); and were subjected to rigorous quality assurance and quality control procedures. Regular ISO 17025 accredited audits encompassed leak checks, flow and microbalance calibrations and ensured measurement traceability to UK national metrological standards. The sites represented a range of locations from the measurements at the kerbside at Marylebone Road, London, to the rural measurements made at East Kilbride outside Glasgow. As such they were deemed fully representative of site and seasonal variations across the UK.

Leckel Kleinfiltergerät PNS-X8 (KFG) samplers were used to collect PM_{10} onto a pre-weighed Emfab filters (Pall Corp., NY, USA; Type: EMFAB TX40HI20-WW; Part No.: 7221) at a flow rate of 2.3 $m^3\ h^{-1}$. Filters were loaded up to 8 days in advance and removed within 1 h of sample changeover; at 10 am or 11 am each day. Filters were then transported to the laboratory under chilled conditions, re-weighed under standardised conditions in a glove box 20 (±1) °C and 50 (±5)% relative humidity after a suitable period of conditioning to determine the mass of particulate collected on the filter. Measurements of sample volume at ambient conditions were used to calculate a mass concentration of PM_{10}. The conditioning and weighing methodology differed from the CEN methodology for PM_{10} (CEN, 1998) and was instead based on the more stringent CEN methodology for $PM_{2.5}$ (CEN, 2003) although stricter protocols for handling and weighing were also used as proposed in Brown et al. (2006). These included chilled storage and transport, additional conditioning time pre- and post-exposure, tighter temperature and relative humidity controls and the re-weighing of unloaded and loaded filters to ensure repeatability (leading to discarding of filters) (Harrison, 2006).

The TEOMs sampled air through a Rupprecht & Patashnick Co.(R&P) PM_{10} inlet at 1 $m^3\ h^{-1}$ and diverted 0.18 $m^3\ h^{-1}$ to the microbalance for mass measurement. The microbalance consisted of a hollow glass tapered tube, clamped at one end and free to oscillate at the other; an exchangeable filter was placed on the free end. The frequency of oscillation was measured and recorded by a microprocessor at 2-s intervals and provided a 15 min running mean concentration. The filter and the air stream passing through it were heated to 50 °C to maintain the filter above the dew point and at a low relative humidity, therefore reducing water uptake (Ruppecht and Patashnick, 1991).

To enable a valid comparison between the different measurement methods, adjustments were made to the TEOM measurements to remove the default post-measurement factors as detailed in Green and Fuller (2006a,b) to calculate mass concentrations at ambient conditions. The first adjustment removes the US EPA defined correction factor in the TEOM (TEOM = 3.0 $\mu g\ m^{-3}$ + 1.03 Raw TEOM), which was included to account for the relative underestimation when compared to the US EPA gravimetric reference method (Ruppecht et al., 1992). The second corrected for the reporting conditions of the TEOM, which default to 25 °C, and 1 atmosphere pressure using the atmospheric temperature and pressure measured by the instrument to calculate mass concentrations at ambient conditions.

Table 1
Summary of measurement site locations. Instrument types are defined as: R = Gravimetric Reference, F = FDMS, T = TEOM.

Site	Type	Lat/Long (Datum: WGS84)	Instruments	Monitoring Period
Acton Town Hall	Roadside	Lat: 51.506586N, Long: 0.268023W	F, T	09/09/05–19/01/06
Belvedere	Suburban	Lat: 51.490685N, Long: 0.159205E	F, T	29/04/04–19/01/06
Birmingham	Urban Background	Lat: 52.455443N, Long: 1.928922W	R, F, T	28/11/04–22/05/05
Bristol	Roadside	Lat: 51.449341N, Long: 2.584642W	R, F, T	13/10/05–19/01/06
East Kilbride	Suburban	Lat: 55.755416N, Long: 4.169038W	R, F, T	13/10/05–12/01/06
Drax Hemingborough	Rural	Lat: 53.762953N, Long: 0.985131W	F	29/05/04–01/01/06
Drax Barlow	Urban Background	Lat: 53.750587N, Long: 1.011211W	F	12/09/04–01/01/06
Marylebone Road	Kerbside	Lat: 51.522393N, Long: 0.154700W	F, T	23/01/04–19/01/06
Millennium Village	Urban Background	Lat: 51.492575N, Long: 0.017756E	T	14/08/04–19/06/06
North Kensington	Urban Background	Lat: 51.520967N, Long: 0.213568W	F, T	01/01/04–19/01/06
Teddington	Suburban	Lat: 51.424331N, Long: 0.345714W	R, F, T	14/11/04–21/03/05
Thames Road	Roadside	Lat: 51.457146N, Long: 0.195279E	F, T	03/04/04–19/01/06
Westhorne Avenue	Roadside	Lat: 51.456350N, Long: 0.040744E	T	21/12/04–19/01/06

The FDMS aims to measure the total mass concentration of airborne particulate matter and quantify the mass changes of the filter due to evaporative and condensation processes that will affect the measurements' concentrations at ambient conditions. It has been shown to measure the total mass of $PM_{2.5}$, including semi-volatile ammonium nitrate and organic material (Grover et al., 2005) and has demonstrated equivalence to the reference measurement in the UK (Harrison, 2006). As discussed, the reference measurement is prone to both positive and negative artefacts from the adsorption of organic gases onto the filter, absorption and evaporation of semi-volatile components and the incomplete removal of particle bound water. Equivalence to this method should therefore be considered with these uncertainties in mind and an increase in the magnitude of any of these effects could result in the FDMS not demonstrating equivalence.

The FDMS was based on TEOM technology; it sampled air through an R&P PM_{10} inlet at 1 m^3 h^{-1} and 0.18 m^3 h^{-1} was diverted to the microbalance. The diverted air passed through a diffusion dryer to remove water from the sample; this allowed the mass to be measured at 30 °C rather than 50 °C. After the dryer, measurement was alternated between two cycles (base and purge), switching between them every 6 min; the change in mass on the filter was measured by the microbalance during both cycles. The FDMS base measurement provided a mass concentration of PM_{10} analogous to that measured by the TEOM; the difference being the dryer and the reduced sampling temperature. During the purge cycle a filter, chilled to 4 °C, removed particulate matter from the sample stream. This purged air was passed through the microbalance filter and the change in mass of filter measured. A total particulate matter concentration measured by the FDMS was calculated as FDMS = FDMS base − FDMS purge. To enable a valid comparison between the measurement methods, adjustments were made to the FDMS (where the instrument configuration required) so that measurements were reported at ambient temperature and pressure as described above.

3. Model derivation

The Volatile Correction Model (VCM) derivation has three steps, which results in a gravimetric reference equivalent concentration denoted as $TEOM_{VCM}$. The FDMS measurement equation (FDMS = FDMS Base − FDMS purge) formed the basis for the model development and Fig. 1 shows how each of the terms was substituted to derive a gravimetric reference equivalent PM_{10} measurement.

1. The FDMS met the equivalence criteria and could therefore be considered equivalent to the method (EC, 2005; Harrison, 2006).
2. The difference between the TEOM and FDMS base measurement reflected the increased volatilisation at the higher sampling temperature of the TEOM (50 °C) compared to the FDMS (30 °C). Mignacca and Stubbs (1999) showed a clear relationship between TEOM sampling temperature (at 30 °C, 40 °C and 50 °C) and mass loss by examining the difference between mass concentrations at different temperatures; the difference between the mass at 30 °C and 50 °C was 2.5 μg m^{-3}; this was similar to the differences experienced in the UK (Green et al., 2007). The difference between the FDMS base and TEOM measurements was well correlated with the FDMS purge measurement in this study. This was most likely due to the incomplete loss of semi-volatile compounds at 30 °C relative to 50 °C. To provide an input for the model, linear regression analysis (forced through zero) of the difference between FDMS base and TEOM daily mean measurements, and the FDMS purge measurements was undertaken for all nine sites operated between 2004 and 2006 and these analyses are shown in Table 2; an example is shown in Fig. 2. Examining these results, there was no consistent site type or regional difference, however, the two locations where low regression slopes were measured could be considered unusual in relation to other sites in the UK. Thames Road experienced substantial construction work very close to the monitoring site during the measurement

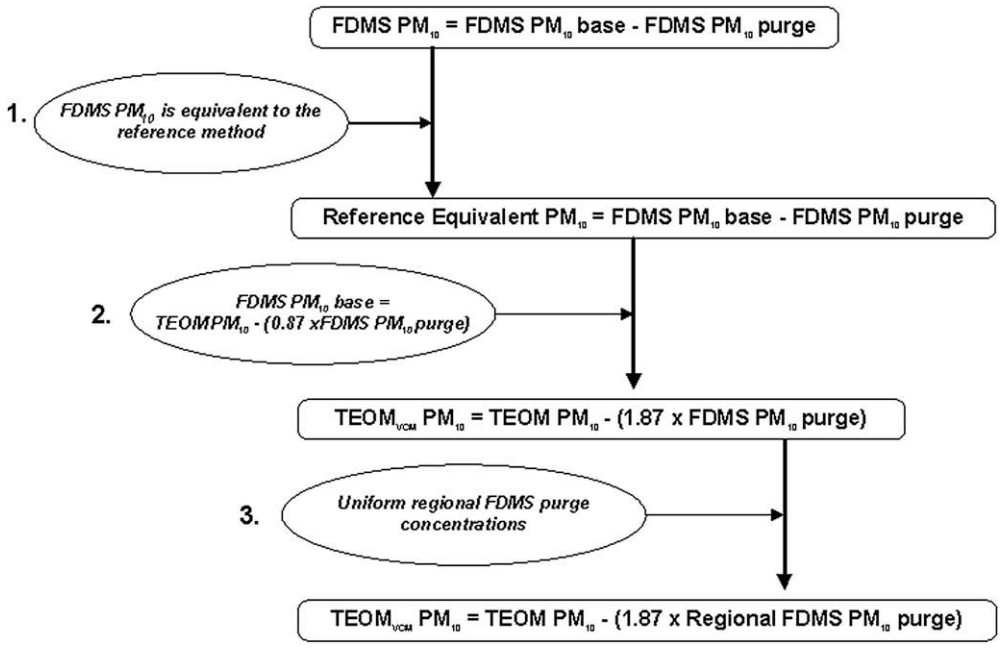


Fig. 1. Stepwise derivation of the Volatile Correction Model.

Table 2
Slope of the linear regression analysis between the FDMS purge measurements (independent variable) and (TEOM–FDMS Base) measurements (dependent variable) from each of the sites. The UK Equivalence Programme sites are shown as the mean of the four potential combinations of the paired instruments (2 TEOMs and 2 FDMS, i.e. TEOM1–FDMS1, TEOM1–FDMS2, TEOM2–FDMS1, TEOM2–FDMS2); the range is shown in brackets. The mean excluded Marylebone Road and Thames Road.

Site	Slope	R^2
Acton Town Hall	1.12	0.81
Belvedere	0.63	0.5
Birmingham	0.98 (0.96–0.99)	0.79 (0.79–0.84)
Bristol	0.78 (0.68–0.88)	0.52 (0.42–0.71)
East Kilbride	0.83 (0.58–1.07)	0.38 (0.15–0.73)
Marylebone Road	0.31	0.44
North Kensington	0.8	0.65
Teddington	1.00 (0.90–1.07)	0.66 (0.56–0.77)
Thames Road	−0.06	−0.28
Mean (excluding Marylebone Road and Thames Road)	0.87	0.61

period. Marylebone Road was very strongly influenced by emissions from the 80,000 vehicles that pass the site every day, it is closer to the road (1.5 m from the kerb) than any of the other sites in this study (Acton Town Hall was 3 m, Bristol 10 m, Thames Road 15 m to main road (7 m to access road) and Westhorne Ave 12 m) and in the AURN and the daily vehicle flow is by far the largest (Acton Town Hall 25,000, Bristol 43,000, Thames Road 31,000 and Westhorne Ave 45,000). Results from these two sites confounded the wider geographical application of the model and they were therefore excluded from the model derivation. The mean linear regression slope calculated excluding the results from Marylebone Road and Thames Road provided a value of 0.87 (including these sites it would have been 0.71) (Table 2). The FDMS base concentration was therefore represented as TEOM − 0.87 FDMS purge. An alternative model parameterisation using the slope and intercept from orthogonal regression analysis was also tested as it was felt that this reflected the relationship of the difference between the FDMS base and the TEOM and the FDMS purge

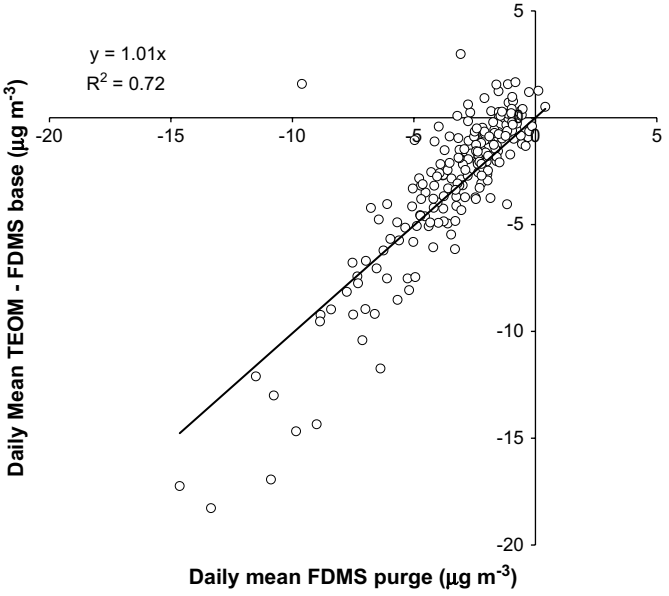


Fig. 2. Correlation between one of the FDMS instrument purge measurements and the (TEOM–FDMS Base) measurement from single FDMS and TEOM instruments at Teddington between November 2004 and July 2005, slope of line of best fit (intercept forced through zero) and r^2 are shown.

more accurately. However, it was found to perform less well than the model using the linear regression parameterisation, the data is therefore not shown.

3. The volatile particulate matter concentrations were found to be similar on a regional scale. This uniformity is demonstrated in Table 3, which shows mean purge concentrations from the FDMS instruments in London, these were separated by distances of up to 34 km. The mean purge concentrations differed by 1.5 µg m^{-3} (between −4.2 µg m^{-3} and −2.8 µg m^{-3}). The large standard deviations of the daily means for the time series demonstrated the large variation in the daily mean purge concentrations. However, despite this the linear regression correlation coefficients (r^2) between the daily means at each site were very high; between 0.74 and 0.98 with an average of 0.92 as shown in Table 3. It was therefore deemed reasonable to assume that a single FDMS purge daily mean measurement was representative of the purge concentration over a wider region. The size of this regional area was determined in the model tests described below.

4. Demonstration of equivalence

The field test procedure for the demonstration of equivalence was determined by the EC Working Group on Guidance for the Demonstration of Equivalence (EC, 2005) and involved four statistical tests for the evaluation of field measurements to determine equivalence between a "candidate" instrument (this could be an automatic monitor or gravimetric analysis carried out on non-reference sampler but in this case the "candidate" was the model) and the gravimetric reference method. All tests were undertaken on daily mean PM_{10} concentrations.

1. For a valid demonstration of equivalence at least four field tests were required, each with paired reference samplers and paired candidate instruments and a minimum of 40 daily mean measurements for both data sets. At least 20% of the daily mean gravimetric reference measurements had to be greater than the daily mean limit value concentration laid down in the EU Directive (50 µg m^{-3}).
2. The concentrations measured by each sampler pair (reference and candidate) deployed during the field tests had to agree within a specified uncertainty. This between sampler uncertainty was a measure of the paired instruments' ability to measure the same daily mean concentration. The EC Working Group stipulated that the between sampler uncertainty for both the gravimetric reference measurements had to be less than 2 µg m^{-3} for the evaluation to proceed and the between sampler uncertainty for the paired candidate instruments had to be less than 3 µg m^{-3}. The paired daily mean concentrations were averaged for use in the remainder of the test procedure.
3. An orthogonal linear regression relationship was calculated between the daily mean concentrations of the reference method and the daily mean concentrations provided by the model. The resulting slopes and intercepts were compared for each of the four field tests, the whole data sets and for the subset of daily mean concentrations greater or equal to 50% of the EU limit values (i.e. 20 µg m^{-3} for annual means and 25 µg m^{-3} for daily means). If the regression slopes were significantly greater or less than 1, then a slope correction should be applied to the measurements from the candidate instrument. If all intercepts were significantly greater or less than zero an intercept correction should be applied. If both these conditions occur the combination of slope and intercepts was applicable. Here we followed the recommendations of

Table 3
Mean FDMS purge concentrations (upper) and correlation matrix of daily mean FDMS purge concentrations (lower) between 1st January 2004 and 31st December 2005. The two Teddington FDMS instruments are differentiated by the instrument serial number.

	Acton Town Hall	Belvedere	Millennium Village	North Kensington	Marylebone Road	Teddington (#24431)	Teddington (#24447)	Thames Road	Westhorne Ave
n (daily means)	106	389	118	461	663	224	186	461	256
Mean μg m^{-3} ± SD	−2.9 ± 2.7	−3.8 ± 2.7	−2.9 ± 1.8	−4.2 ± 3.1	−4.0 ± 2.8	−3.6 ± 2.8	−3.4 ± 2.8	−3.4 ± 2.8	−2.8 ± 3.1
Correlation Coefficients (daily means)									
Acton Town Hall	1.00								
Belvedere		1.00							
Millennium Village		0.91	1.00						
North Kensington	0.92	0.92	0.74	1.00					
Marylebone Road	0.87	0.95	0.93	0.91	1.00				
Teddington (#24431)		0.96	0.95	0.90	0.92	1.00			
Teddington (#24447)		0.96	0.97	0.85	0.91	0.98	1.00		
Thames Road	0.90	0.92	0.91	0.89	0.90	0.92	0.93	1.00	
Westhorne Ave	0.96	0.98	0.84	0.96	0.95	0.96	0.97	0.92	1.00

Harrison (2006); that corrections only need be applied where *all* significant slopes were greater or less than 1 or/and *all* significant intercepts were greater or less than zero.

4. The combined relative expanded uncertainty at the Limit Value (W_{CM}) was calculated. This is a function of the sum of the relative residuals from the orthogonal regression, the concentration from the original regression equation at the Limit Value and the between sampler uncertainty of the gravimetric reference method. The W_{CM} was calculated for each field test, the whole data set and for the subset of daily mean concentrations greater or equal to 50% of the upper assessment threshold in the Daughter Directive. The maximum value of W_{CM} had to be less than the expanded uncertainty data quality objective of 25% stipulated in the Daughter Directive.

Here we again followed the recommendations of Harrison (2006) and departed from the Guidance in two respects. Firstly, that the candidate was not deemed to fail the equivalence test on the sole basis of W_{CM} exceeding 25% for the data set of measurements that was less than 50% of the Limit Value. Secondly, due to the low PM_{10} concentrations measured at a rural location at East Kilbride in Scotland, the candidate was not deemed to fail the equivalence test on the sole basis of this deployment.

The model was then tested against the equivalence criteria in two experiments. Experiment 1 was designed to test the equivalence of the model excluding regional variation in volatile particulate matter. This used measurements from collocated pairs of reference samplers and FDMS and TEOM instruments available from Birmingham, Bristol, East Kilbride and Teddington. If we considered each site as having two TEOMs (A and B) and two FDMS (C and D) we had eight possible individual modelled 24 h mean time series at each site four possible pairs (1–4) of modelled daily mean time series to test at each site:

Modelled PM_{10} 1a = TEOM A − 1.87 FDMS purge C
Modelled PM_{10} 1b = TEOM B − 1.87 FDMS purge C
Modelled PM_{10} 2a = TEOM A − 1.87 FDMS purge D
Modelled PM_{10} 2b = TEOM B − 1.87 FDMS purge D
Modelled PM_{10} 3a = TEOM A − 1.87 FDMS purge C
Modelled PM_{10} 3b = TEOM B − 1.87 FDMS purge D
Modelled PM_{10} 4a = TEOM A − 1.87 FDMS purge D
Modelled PM_{10} 4b = TEOM B − 1.87 FDMS purge C

The mean of each pair of modelled PM_{10} 24 h mean time series was compared to the reference measurements and subjected to the full equivalence test criteria.

Experiment 2 was designed to test the equivalence of the model, including the regional variation in volatile particulate matter. This experiment used the collocated gravimetric reference method and TEOM measurements from Birmingham, Bristol, East Kilbride and Teddington and FDMS purge measurements from another location. Two TEOMs (A and B) were used to create two modelled 24 h mean time series using the purge measurements from all other non-collocated FDMS sites (Z) using the model equation:

$TEOM_{VCM}$ = TEOM A − 1.87 FDMS purge Z
$TEOM_{VCM}$ = TEOM B − 1.87 FDMS purge Z

The mean of the two modelled PM_{10} 24 h mean time series was then subjected to the full equivalence test criteria. To aid the determination of the spatial applicability of the model the distance between each FDMS site (Z) and the equivalence site was also calculated.

5. Results

The performance of the model in experiment 1 is summarised in Table 4. For completeness, data sets with less than 40 data pairs are reported, however, these results should be treated with caution. The between sampler uncertainty for the model (0.88 μg m^{-3}) was less than 3 μg m^{-3}, and as such, the model was considered suitable as a candidate method. This was lower than the between sampler uncertainty of 1.12 μg m^{-3} reported for the FDMS but higher than the TEOM value, on which the model is based, of 0.52 μg m^{-3} (Harrison, 2006). All significant slopes of the individual and combined data sets were both greater and less than 1 (winter range: 0.84–1.26, summer range: 0.93–1.06), and all significant intercepts were both greater and less than zero (winter range: −1.05 to 3.37, summer range: −0.21 to 4.50). As such, the expanded uncertainty could be calculated without the need for any correction for slope or intercept. The expanded uncertainty (W_{CM}) was less than 25% for all but four combinations at East Kilbride in the summer. In line with the recommendations of Harrison (2006) the candidate was not deemed to fail the equivalence test on the basis of the data set of measurements <50% of the Limit Value or the individual deployments East Kilbride. This reflects the problems associated with regression calculations where there is significant scatter on data that are restricted to within a narrow range (i.e. low PM_{10} concentrations). The W_{CM} of some of these were lower than those reported by the FDMS. This reflects the accuracy of the 0.87 factor in the model at collocated sites, however, it should be remembered that the model was in part derived from these measurements and a good agreement was expected.

The performance of the model in experiment 2 is summarised in Table 5; only data sets with 40 or more data pairs were included. The maximum 24-h between sampler uncertainty (0.89 μg m^{-3}) was less than 3 μg m^{-3}, this low value was a result of the model

Table 4
Summary of experiment 1, the comparison between model and gravimetric reference method without geographical variation in volatile PM_{10}. The pair of modelled time series used is denoted by the FDMS serial numbers and season. Summer (S) 1st April–30th September, Winter (W) 1st October–31st March.

Data set (Serial #1, Serial #2, Season)	n	Distance between sites (km)	Between sampler uncertainty ($\mu g\ m^{-3}$)	r^2	Slope (b) $\pm U_b$	Intercept (a) $\pm U_a$	W_{CM} (%)	
							Annual Limit Value	Daily Limit Value
Birmingham (04443,04443,S)[b]	39	0	0.38	0.96	0.94 ± 0.03	2.00 ± 0.64	9	8
Birmingham (04443,25053,S)[b]	39	0	1.03	0.96	0.95 ± 0.03	2.69 ± 0.66	10	7
Birmingham (25053,04443,S)[b]	39	0	1.33	0.96	0.95 ± 0.03	2.69 ± 0.66	10	7
Birmingham (25053,25053,S)	41	0	0.38	0.95	0.95 ± 0.04	3.58 ± 0.73	14	10
Birmingham (04443,04443,W)	47	0	0.15	0.91	0.96 ± 0.04	0.20 ± 0.87	12	11
Birmingham (04443,25053,W)	47	0	0.81	0.91	0.96 ± 0.04	−0.12 ± 0.85	13	11
Birmingham (25053,04443,W)	47	0	0.82	0.91	0.96 ± 0.04	−0.12 ± 0.85	13	11
Birmingham (25053,25053,W)	59	0	0.17	0.95	0.95 ± 0.03	−0.21 ± 0.67	15	14
Bristol (24431,24431,S)[b]	36	0	0.38	0.96	1.04 ± 0.04	2.01 ± 0.90	21	19
Bristol (24431,24447,S)[b]	36	0	1.12	0.96	1.05 ± 0.04	1.08 ± 0.88	19	17
Bristol (24447,24431,S)[b]	36	0	1.24	0.96	1.05 ± 0.04	1.08 ± 0.88	19	17
Bristol (24447,24447,S)[b]	36	0	0.38	0.96	1.06 ± 0.04	0.12 ± 0.88	16	15
Bristol (24431,24431,W)	55	0	0.53	0.95	0.93 ± 0.03	4.50 ± 0.69	14	10
Bristol (24431,24447,W)	55	0	1.10	0.96	0.93 ± 0.03	3.89 ± 0.66	12	9
Bristol (24447,24431,W)	55	0	1.00	0.96	0.93 ± 0.03	3.89 ± 0.66	12	9
Bristol (24447,24447,W)	57	0	0.54	0.96	0.93 ± 0.02	3.31 ± 0.63	11	8
East Kilbride (25053,25053,S)[a,b]	36	0	0.27	0.86	1.26 ± 0.08	1.43 ± 0.75	59	57
East Kilbride (25053,04443,S)[a,b]	33	0	1.59	0.79	1.21 ± 0.10	1.01 ± 0.85	48	47
East Kilbride (04443,25053,S)[a,b]	33	0	1.26	0.79	1.21 ± 0.10	1.01 ± 0.85	48	47
East Kilbride (04443,04443,S)[a,b]	33	0	0.24	0.76	1.25 ± 0.11	−0.07 ± 0.93	50	50
East Kilbride (25053,25053,W)	54	0	0.61	0.91	0.98 ± 0.04	3.37 ± 0.50	15	12
East Kilbride (25053,04443,W)	54	0	1.97	0.93	0.97 ± 0.04	2.55 ± 0.44	9	7
East Kilbride (04443,25053,W)	54	0	1.16	0.93	0.97 ± 0.04	2.55 ± 0.44	9	7
East Kilbride (04443,04443,W)	54	0	0.61	0.93	0.97 ± 0.04	1.62 ± 0.44	7	5
Teddington (24431,24431,S)	50	0	0.77	0.89	0.89 ± 0.04	2.09 ± 0.97	19	18
Teddington (24431,24447,S)	49	0	1.03	0.90	0.87 ± 0.04	2.21 ± 0.94	21	21
Teddington (24447,24431,S)	49	0	1.41	0.90	0.87 ± 0.04	2.21 ± 0.94	21	21
Teddington (24447,24447,S)	50	0	0.88	0.89	0.84 ± 0.04	2.49 ± 0.94	25	25
Teddington (24431,24431,W)[b]	29	0	0.32	0.95	0.90 ± 0.04	−0.33 ± 1.01	25	24
Teddington (24431,24447,W)[b]	20	0	0.64	0.96	0.93 ± 0.04	−1.05 ± 1.21	22	20
Teddington (24447,24431,W)[b]	20	0	0.52	0.96	0.93 ± 0.04	−1.05 ± 1.21	22	20
Teddington (24447,24447,W)[b]	20	0	0.35	0.96	0.92 ± 0.04	−1.05 ± 1.26	23	21
All Data	1362	0	0.88	0.92	0.93 ± 0.01	2.33 ± 0.15	13	11
Data $<20\ \mu g\ m^{-3}$	970	0	0.88	0.70	0.99 ± 0.02	1.68 ± 0.23	12	–
Data $\geq 20\ \mu g\ m^{-3}$	392	0	0.93	0.85	0.95 ± 0.02	1.34 ± 0.62	18	–
Data $<25\ \mu g\ m^{-3}$	1101	0	0.87	0.77	1.01 ± 0.01	1.36 ± 0.21	–	12
Data $\geq 25\ \mu g\ m^{-3}$	261	0	0.91	0.82	0.99 ± 0.03	−0.60 ± 0.99	–	16

[a] Indicates data sets with W_{CM} greater than 25%.
[b] Indicates data sets with less than 40 paired measurements.

being based on the TEOM measurements that had a between sampler uncertainty of 0.52 $\mu g\ m^{-3}$. The model was suitable for consideration as a candidate method. All significant slopes of the individual data sets were both greater and less than 1 (range: 0.67–1.29), and all significant intercepts were both greater and less than zero (range: −0.21 to 8.26). As such, the expanded uncertainty could be calculated without the need for any correction for slope or intercept. The expanded uncertainty (W_{CM}) was less than 25% for all but ten combinations. The effect of distance on the W_{CM} is shown in Fig. 3. It is clear that at distances of up to 200 km the expanded uncertainty of the modelled concentration remained below the 25% for both the annual and the daily limit values (with the exception of Teddington to Westhorne Avenue during the summer). The W_{CM} for this data pair was 28% for both the annual and daily limit values. This was considered a marginal breach of the 25% limit for W_{CM} given that the overwhelming majority of deployments (22 out of 23) with separation distances of less than 200 km met the criteria. A closer examination of the regression statistics for this combination shows a good r^2 (0.91), however, the relatively small slope (0.86) and intercept (0.55) led the model to an underestimate the mean gravimetric reference method concentration by 2 $\mu g\ m^{-3}$. The intercept close to zero was the result of the purge measurements from this site being close to zero during the periods of low particulate volatility during the summer. This single marginal failure was therefore not deemed suitable justification for rejection and the model was therefore deemed to meet the criteria for the reference equivalence using FDMS purge measurements from remote sites up to approximately 200 km distant.

6. Discussion

It is necessary to consider the physical and chemical basis of the model to understand the validity of this correction methodology. In particular, the use of semi-volatile particulate matter (made during the purge measurement cycle) from regionally deployed FDMS instruments and the empirical relationship derived from collocated TEOM and FDMS base measurements.

6.1. Purge measurements

Purge measurements are influenced by the absorption and desorption processes on the filter; the relative effect of these will depend on the following:

1. The concentration of semi-volatile components and gases passing over the filter during the purge cycle of the FDMS.
2. The composition of the semi-volatile particulate deposited and therefore loss of volatile material from the filter.
3. The absorbing mass of particulate already deposited on the filter, as well as its ability to absorb gaseous phase components

Table 5
Summary of experiment 2, the comparison between model and gravimetric reference method with geographical variation in volatile PM_{10} sorted by increasing distance. The pair of modelled time series used is denoted by the FDMS serial numbers and season. Summer (S) 1st April–30th September, Winter (W) 1st October–31st March.

Data set Equivalence Site (FDMS (Serial # where applicable), Season)	n	Distance between sites (km)	Between sampler uncertainty ($\mu g\ m^{-3}$)	r^2	Slope (b) ± U_b	Intercept (a) ± U_a	W_{CM} (%)	
							Annual Limit Value	Daily Limit Value
Teddington (North Kensington, S)	48	14	0.89	0.84	0.88 ± 0.05	3.53 ± 1.22	21	19
Teddington (Marylebone Road, S)	52	17	0.88	0.88	0.89 ± 0.04	3.42 ± 1.02	17	15
Teddington (Westhorne Ave, S)[a]	44	27	0.35	0.91	0.86 ± 0.04	0.55 ± 0.93	29	28
Teddington (Belvedere, S)	44	36	0.86	0.91	0.91 ± 0.04	3.81 ± 1.03	15	13
Birmingham (Drax Hemingborough, W)	54	156	0.21	0.83	1.02 ± 0.06	−0.68 ± 1.24	19	15
Birmingham (North Kensington, S)	42	157	0.40	0.86	0.83 ± 0.05	5.94 ± 1.06	18	18
Birmingham (Teddington 24431,W)	55	158	0.21	0.83	0.90 ± 0.05	2.02 ± 1.11	19	18
Birmingham (Teddington 24447,W)	43	158	0.18	0.88	0.83 ± 0.05	3.02 ± 1.03	22	24
Teddington (Birmingham 04443,S)	46	158	0.44	0.84	0.96 ± 0.06	0.90 ± 1.37	22	18
Teddington (Birmingham 25053,S)	47	158	0.83	0.83	0.99 ± 0.06	1.78 ± 1.44	24	19
Birmingham (Drax Hemingborough, W)	41	159	0.21	0.86	0.98 ± 0.06	−1.01 ± 1.21	17	14
Birmingham (Marylebone Road, S)	42	160	0.38	0.93	0.85 ± 0.04	5.39 ± 0.74	12	13
Birmingham (Marylebone Road, W)	56	160	0.21	0.83	0.95 ± 0.05	2.31 ± 1.19	18	14
Bristol (Acton Town Hall, W)	54	161	0.52	0.95	0.90 ± 0.03	2.80 ± 0.74	13	13
Bristol (Marylebone Road, W)	55	169	0.52	0.92	0.89 ± 0.03	5.00 ± 0.90	15	12
Birmingham (Westhorne Ave, W)	40	175	0.21	0.84	0.92 ± 0.06	0.25 ± 1.24	21	19
Birmingham (Belvedere, S)	45	179	0.38	0.91	0.87 ± 0.04	5.83 ± 0.85	14	11
Birmingham (Belvedere, W)	54	179	0.21	0.86	0.87 ± 0.05	3.58 ± 1.02	17	17
Bristol (Westhorne Ave, W)	53	182	0.54	0.95	0.88 ± 0.03	3.26 ± 0.76	15	15
Birmingham (Thames Road, W)	56	183	0.21	0.82	0.99 ± 0.06	0.69 ± 1.33	21	17
Bristol (Thames Road, W)	47	193	0.53	0.92	0.85 ± 0.04	3.46 ± 0.96	19	19
Teddington (Drax Barlow, S)[a]	46	263	0.92	0.66	0.58 ± 0.06	6.87 ± 1.34	54	60
Bristol (Drax Hemingborough, W)[a]	56	280	0.51	0.83	0.86 ± 0.05	9.57 ± 1.27	29	20
East Kilbride (Drax Hemingborough, W)[a]	50	302	0.60	0.65	1.08 ± 0.09	7.10 ± 1.11	55	47
Bristol (East Kilbride 25053,W)[a]	56	490	0.54	0.92	0.67 ± 0.03	8.26 ± 0.68	28	35
Bristol (East Kilbride 04443,W)[a]	56	490	0.54	0.93	0.68 ± 0.02	6.02 ± 0.64	36	42
East Kilbride (Bristol 24431,W)[a]	51	490	0.61	0.66	1.29 ± 0.10	3.01 ± 1.26	77	72
East Kilbride (Bristol 24447,W)[a]	49	490	0.61	0.72	1.22 ± 0.09	2.24 ± 1.12	58	55
East Kilbride (Acton Town Hall, W)[a]	52	538	0.6	0.67	1.15 ± 0.09	1.75 ± 1.08	44	40
East Kilbride (Marylebone Road, W)[a]	51	540	0.62	0.61	1.06 ± 0.09	4.52 ± 1.17	40	34
East Kilbride (Westhorne Ave, W)[a]	52	553	0.61	0.73	1.09 ± 0.08	1.98 ± 0.97	33	30
East Kilbride (Thames Road, W)	44	558	0.60	0.62	1.00 ± 0.09	2.57 ± 1.17	24	19

[a] Indicates data sets with expanded uncertainties greater than 25%.

(organic aerosol to into which semi-volatiles can partition, aqueous aerosol in the case of water soluble organics and hygroscopic aerosol in the case of water vapour).

During the purge measurement cycle, the mass lost due to evaporation of semi-volatile material tended to exceed the mass gained due to any condensation of gaseous material onto the filter. This resulted in a negative FDMS purge measurement in most cases, although it is accepted that both absorption and evaporation occurred at the same time.

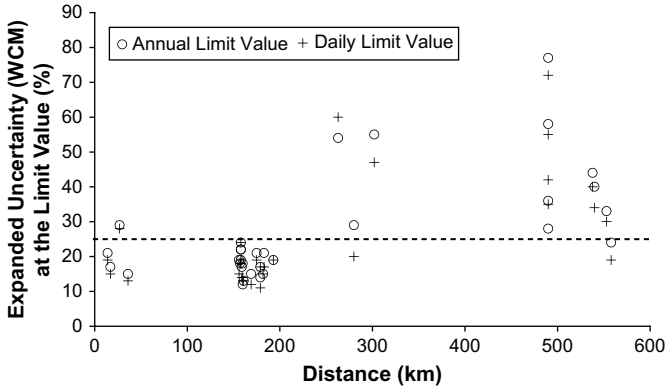


Fig. 3. The relationship between W_{CM} for the daily and annual mean limit values with distance for the comparison between model and the gravimetric reference method. The data quality objective of 25% is shown as a dotted line.

Absorption processes during the purge cycle of the FDMS were minimised, firstly by the chilled purge filter that condensed gaseous phase aerosol (such as organic compounds), and secondly by the diffusion dryer that reduced the relative humidity of the sample stream below the ambient dew point. Nevertheless, 8% of the daily mean FDMS purge concentrations were positive, indicating that absorption was occurring during certain conditions. These positive measurements tended to occur during periods of low PM_{10} mass concentration, and may have been responsible for the predominance of positive intercepts shown in Table 5. If we assume that the purge filter and the diffusion dryer are effective in minimising the impact of organic aerosol and the condensation of water respectively, then the absorption of water vapour by hygroscopic aerosol is the most likely cause of these positive measurements.

The influence of water condensation on the FDMS and TEOM filters was expected to be minimal, as the dew point at both the TEOM and FDMS measurement filters remained low due to heating of the sample in both the TEOM and FDMS instruments and the additional drying of the sample in the FDMS.

The composition of the semi-volatile particulate deposited has been shown to influence the FDMS purge measurement (Grover et al., 2005; Favez et al., 2007) and the vaporisation from the differential TEOM (Hering et al., 2004; Jaques et al., 2004; Lee et al., 2005). Ammonium nitrate, in particular, has shown an excellent agreement with the vaporisation measurements in some studies (Green, 2004; Grover et al., 2005; Hering et al., 2004; Favez et al., 2007) and less so in others (Jaques et al., 2004). The role of SVOC is less clear as measurements are more challenging. Consequently,

a range of an organic metrics such as OC (Lee et al., 2005), SVOC (Grover et al., 2005) and UV aethalometry (Favez et al., 2007) were used to demonstrate the effects of SVOC in these studies.

Measurements of semi-volatile organic material were not available in this study, however, it was possible to estimate the source strength of the semi-volatile organic material by examining the concentration of secondary organic aerosol (SOA). SOA is formed through the gas phase oxidation of anthropogenic (e.g. aromatics, alkenes, alkanes and cycloalkanes) and biogenic sources (e.g. terpenes), the products then form aerosol through heterogeneous or homogeneous nucleation (Turpin et al., 2000). The anthropogenic sources (e.g. diesel combustion) and many of the gas-phase oxidants involved in the reactions (e.g. OH) are the same as those responsible for the formation of ammonium nitrate. When correlating the nitrate and SOA measurements from Birmingham, Harrison and Yin (2008) concluded that the main factors influencing secondary organic carbon concentrations (regional transport and loss of the semi-volatile material from the aerosol) are similar to those which determine fine particulate nitrate.

During a short period at the end of 2006, collocated measurements of PM_{10} (measured using the FDMS), elemental and organic carbon concentration in PM_{10} (measured using a Sunset Laboratories Carbon Aerosol Analysis Lab Instrument; Butterfield et al., 2007) and nitrate in $PM_{2.5}$ (measured using a Thermo 8400N; Hayman et al., 2008) were made at Marylebone Road. The large primary organic emissions at this location meant that it was not the ideal data set for this type of analysis, however, these were the only collocated measurements available in the UK.

The SOA concentrations were calculated using the method described by Turpin and Huntzicker (1990) using the following equation:

$$\mathrm{SOA} = \mathrm{OC} - \left(\frac{\mathrm{OC}}{\mathrm{EC}}\right)_{\mathrm{prim}} \times \mathrm{EC}$$

where $(OC/EC)_{prim}$ was the minimum OC:EC ratio for the measurement period; in this case 0.34 (close to the Birmingham roadside OC:EC ratio of 0.4 from Harrison and Yin, 2008). To account for associated oxygen and hydrogen atoms in the organic compounds, the measured organic carbon mass of SOA was multiplied by 2.1, this is the value recommended for rural aerosol by Turpin and Lim (2001) and was considered here to best represent SOA from long range sources. The $PM_{2.5}$ nitrate was assumed to be present as NH_4NO_3 and therefore multiplied by 1.29 to give a mass of NH_4NO_3. The resulting concentrations of SOA, ammonium nitrate and PM_{10} FDMS purge measurements are shown in Fig. 4. This demonstrates that the concentrations of ammonium nitrate and SOA are well correlated with the FDMS purge measurements (RMA correlation coefficients (R) of −0.66 and −0.74 respectively). However, the SOA concentration was greater than the concentration of semi-volatile organic material that was affecting the FDMS purge concentrations. This was due to the loss of semi-volatile organic material which would have been influenced by a number of factors, including the ambient and sampling temperatures, pressure drop across the filter, aerosol water concentration, particle concentration, particle surface area, particle organic matter concentration and gas concentration (Turpin et al., 2000). Most important in this case was the difference in sample temperature between the FDMS (30 °C) and the Partisol (ambient), which would have resulted in the loss of semi-volatile organic material.

The influence that any absorbing mass of particulate already deposited on the filter has on the purge measurements is difficult to quantify especially with the respect to absorption of ammonium nitrate and organic gases. However, scanning electron microscopy work has identified influences from the evaporation of particle bound water after collection (Jones et al., 2001).

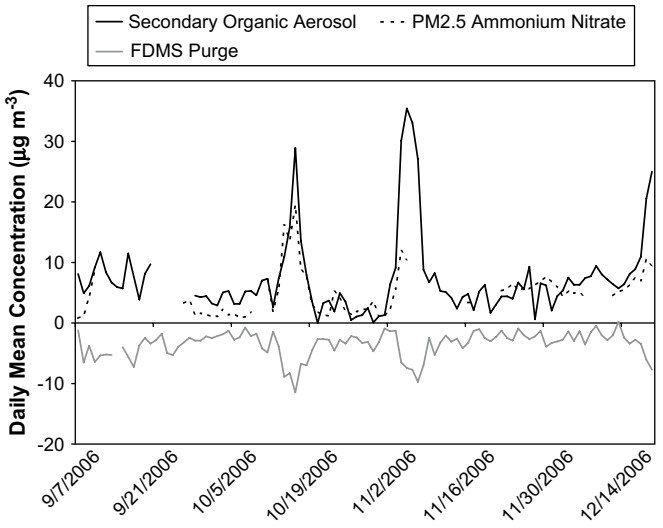


Fig. 4. Time series of daily mean secondary organic aerosol (SOA), ammonium nitrate in $PM_{2.5}$ and PM_{10} FDMS purge measurements made at Marylebone Road between 7th September and 14th December 2006.

The uniformity of the FDMS purge measurements between roadside and background sites demonstrated in Table 3 was surprising given the semi-volatile nature of primary organic emissions (Robinson et al., 2007). However, the 30 °C sampling temperature of the FDMS may lead to the partition of these components into the gaseous phase.

6.2. The relationship between TEOM and FDMS base measurements

The relationship derived between collocated TEOM mass concentrations and FDMS base measurements was also fundamental to the model. Differences between the FDMS base measurement and the TEOM measurement methodology existed in the sampling regime. There were three differences between the instruments in this respect:

1. The FDMS switched between the base and purge measurement cycles every 6 min. It therefore only sampled ambient air for the base measurement 50% of the time.
2. The FDMS used a diffusion drier in the sample stream to remove particle bound water.
3. The FDMS heated the sample stream to 30 °C, rather than the 50 °C used by the TEOM, to provide a stable temperature environment for the microbalance.

Over short time periods and in environments with rapidly changing PM_{10} concentrations, such as close to major roads, this switching may have induced noise in this relationship. However, over longer time periods (daily mean) and at the urban locations used in this study the 50% sampling time of the FDMS should represent ambient concentrations.

The diffusion dryer has the potential to influence the chemical and physical characteristics of the sample. As well as removing water, the dryer removes other compounds such as alcohols, ammonia and amines. It can also convert carbonyl groups (–C═O) into hydroxyl groups through acid catalysis enolization and removed the compound in this way (PermaPureLLC, 2006). Furthermore, some attenuation of particles is recognised to occur in

the dryer due to diffusion losses (small particles) and impaction (large particles) (Dick et al., 1995). The impaction of large particles may have been responsible for some of the differences demonstrated at Marylebone Road and Thames Road, which have a higher concentration of coarse particles (PM_{10}–$PM_{2.5}$) than other sites due to their kerbside and construction influenced environments respectively.

The major difference between the FDMS and TEOM instruments (except Marylebone Road and Thames Road) that affected mass concentrations would therefore have been the difference in sampling temperature. Semi-volatile particulate matter that was volatile at 50 °C but not at 30 °C would have been measured by the FDMS but not by the TEOM; these are likely to be the same components identified as influencing the FDMS purge measurement and the two are consequently correlated (as shown in Fig. 2).

Grover et al. (2005) also examined the difference between the TEOM and the total FDMS mass measurements alongside measurements of the ammonium nitrate and semi-volatile organic material. They found that the difference was mostly explained by ammonium nitrate measurements (as it is in this model), the remainder was accounted for (but in many samples shown to be less than) the measured semi-volatile organic material.

In this study the purge measurement was demonstrated to represent the ammonium nitrate concentration, the difference between the TEOM and the FDMS base measurements (described in the model as 0.87 times the purge measurement) may therefore correspond to the semi-volatile organic material measured by the FDMS base at 30 °C but not by the TEOM at 50 °C due to the gas–particle partitioning effects described earlier. In practice, the division between the ammonium nitrate and semi-volatile organic material is unlikely to be that precise. The model here succeeded in describing the total mass difference between the FDMS and TEOM as a function of the purge measurement because a large fraction of the semi-volatile organic material is secondary in origin and influenced by the same factors that determine ammonium nitrate concentrations (Turpin and Lim, 2001; Harrison and Yin, 2008).

7. Conclusion

The VCM allows the correction of TEOM measurements to enable them to meet the equivalence criteria for the measurement of PM_{10}. This allows the correction of many TEOM instruments by a single FDMS measurement. The model equation is:

$$TEOM_{VCM} = TEOM - 1.87\ FDMS\ purge$$

The model passed the field test procedure for the demonstration of equivalence as described by the EC Working Group on Guidance for the Demonstration of Equivalence (EC, 2005) with departures previously used in the UK submission to the EC (Harrison, 2006). The model could therefore be considered an equivalent method.

The model used measurements of semi-volatile particulate matter made by the FDMS. Importantly, the FDMS measurements may be made at a location some distance away from the TEOM as the semi-volatile particulate matter is uniform over a wide area. The model passed the equivalence criteria using FDMS purge measurements over a maximum distance from the associated TEOM of approximately 200 km. Twenty-two out of 23 tests at less than 200 km passed the equivalence criteria and the single failure was marginal. These findings indicate that the model is a viable tool for correcting measurements from TEOM instruments in national and local government networks using FDMS purge measurements from a more limited network of sites. The model therefore offers large practical and financial benefits for these networks.

Nevertheless, the chemical constituents of the purge measurements and the reasons for difference between the FDMS base measurement and the TEOM measurements still require further examination. Measurements of the semi-volatile organic material and ammonium nitrate are key to providing a greater understanding of these differences.

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